Extra Dimensions and Gravitational Decay

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In this paper, we will discuss how the gravitational decay of cosmic superstrings can help determine the dimension of spacetime. To gain context in the subject, we present intuition for the nature of extra dimensions, explore areas of general relativity and string theory, and carry out important derivations along the way.

I. INTRODUCTION

In his 1916 theory of General Relativity, Albert Einstein theorized the existence of *gravitational waves*, which are waves that propagate through spacetime causing minute compressions and expansions in matter. These are often consequences of large celestial bodies colliding and emitting gravitational radiation. A prime example of such an event is two black holes colliding. Indeed, an event of this type caused the waves that LIGO detected in 2015. While these events release massive amounts of energy, it is dissipated rapidly, making the observed amplitude extremely small, roughly of the order 10^{-26}.

In 2008, there was a large push towards research on cosmic string networks and gravitational waves, but this was mostly theoretical and computational in nature. Additionally, there was (and still is) much talk of the LISA detector, a gravitational wave detector in orbit set to launch in the 2030s, and how it can improve current detection limits. This was obviously before the LIGO detection occurred, so the tone of the research has changed slightly since then. One major takeaway from this era of research and the overarching goal of this paper is to understand the stochastic gravitational wave background from cosmic string decay.

A. Outline

The organization of this paper is as follows: First, we will discuss the nature of small extra dimensions and how they may appear in the observable spacetime. Next, we will discuss gravitational waves in a General Relativistic sense in order to understand how cosmic strings decay and emit gravitational radiation. We then derive the critical dimension $D = 26$ for quantum bosonic strings as one potential answer to the question "How many dimensions does spacetime have?". Attempting to find a general answer, we discuss cosmic superstrings and their decay, using the interaction between gravitational radiation and the number of extra dimensions to point towards future research on this subject.

II. EXTRA DIMENSIONS

To give intuition on what extra dimensions could look like in the observable spacetime, we use the model of small quotient spaces on top of a plane. We first define the term quotient space.

Definition 1. Let V be a vector space over a field $\mathbb{F}$ with a subspace $W \leq V$. Then, for $v, u \in V$, we define an equivalence relation $v \sim u$ if $v - u \in W$. Then, we may find the equivalence class $[v] = \{u + w \mid w \in W\}$ and define the *quotient space* V/W to be the set of equivalence classes under the relation $\sim$, namely

$$V/W := \{[v] \mid v \in V\}. \tag{1}$$

Remark 2. While this definition may seem like overkill, it is actually precise because if two equivalence classes coincide, e.g. $[v] = [u]$ for $v \neq u \in V$, then the set theoretic nature of the definition omits multiplicities, admitting only one representative equivalence class. ♣

We briefly examine some familiar geometric examples of quotient spaces. For more information, see the section on Orbifolds in [8].

Consider the vector space $\mathbb{R}$ and the subspace $\mathbb{Z}$. We find the quotient space $\mathbb{R}/\mathbb{Z}$ to be the set of $x \in [0, 1)$ with the property that $[0] = [1]$, so $0 \sim 1$. We call this an *identification*, or that 0 is *identified with* 1. Geometrically, this forms a 1-dimensional circle as we take the

unit interval and attach the points 0 and 1. As well, for any constant k, we may define the quotient space $\mathbb{R}/k\mathbb{Z}$ by the set of $x \in [0, k)$ with the property that $0 \sim k$. This is the circle with circumfrence k.

It is also useful to define a quotient space by how different elements in the set are identified. For example, the quotient space $\mathbb{R}/2\pi R\mathbb{Z}$ is equivalent to the identification $x \sim x + 2\pi R$. This forms the circle of radius R.

Also, if we consider the pair of identifications in the plane

$$\begin{cases} (x, y) \sim (x + 1, y) \\ (x, y) \sim (x, y + 1), \end{cases}$$

the first identification constructs an open cylinder with circumference 1, and the second identification applied to the cylinder creates a torus. I will denote this as $\mathbb{R}^2/\mathbb{Z}_x/\mathbb{Z}_y =: \mathbb{R}^2/\mathbb{Z}^2$.

To visualize what these extra dimensions could look like, imagine living in $\mathbb{R}^2 \times (\mathbb{R}/\epsilon\mathbb{Z})$, where $\epsilon \ll 1$ is small. To the average observer, this looks like the plane because when on this space, the amount they could move perpendicular to the plane would be negligible, making them think they are restricted to 2 dimensions. $\mathbb{R}^2 \times (\mathbb{R}^2/\epsilon\mathbb{Z}^2)$ works in the same way because instead of a small loop on top of the plane, it is a small torus, and feels much the same because of how small ϵ is, making it virtually undetectable.

This is the general idea of how extra dimensions would appear in our world: small, undetectible extensions of the current space. Determining methods for detecting these types of extra dimensions and rationale for these even existing in the first place is the goal of this paper.

III. GRAVITATIONAL WAVES

We begin this section with a string of definitions that build upon each other in the lead-up to Einstein's Equation. Note: we will let $\partial_\mu = \frac{\partial}{\partial x^\mu}$ and $\partial_{\mu\nu} = \frac{\partial^2}{\partial x^\mu \partial x^\nu}$.

Definition 3.

1. We define the *d'Alembertian operator* $\Box := \partial_\mu \partial^\mu = -\frac{1}{c^2}\partial_{tt} + \Delta$, where Δ is the Laplacian.

2. We define the *Metric Tensor* $g_{\mu\nu}$ as $g_{\mu\nu} := \eta_{\mu\nu} + h_{\mu\nu}$, where

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{2}$$

and $h_{\mu\nu}$ is a perturbation tensor. Additionally, we define $g := \det g_{\mu\nu}$.

3. We define the *Christoffel Symbol* ${\Gamma^\mu}_{\alpha\beta}$ as

$${\Gamma^\mu}_{\alpha\beta} := \frac{1}{2} g^{\mu\nu} \left(\partial_\beta g_{\nu\alpha} + \partial_\alpha g_{\beta\nu} - \partial_\nu g_{\alpha\beta} \right). \tag{3}$$

4. We define the *Riemann Curvature Tensor* as

$${\mathcal{R}^\mu}_{\gamma\alpha\nu} := \partial_\nu {\Gamma^\mu}_{\alpha\gamma} - \partial_\alpha {\Gamma^\mu}_{\nu\gamma} + {\Gamma^\mu}_{\nu\beta}{\Gamma^\beta}_{\alpha\gamma} - {\Gamma^\mu}_{\alpha\beta}{\Gamma^\beta}_{\nu\gamma}. \tag{4}$$

5. We define the *Ricci Tensor* as $R_{\alpha\beta} := {\mathcal{R}^\mu}_{\alpha\mu\beta}$. Additionally, we define the *Ricci Scalar* as $R := {R^\beta}_\beta = g^{\mu\nu} R_{\mu\nu}$.

6. We define the *Einstein Tensor* $G_{\mu\nu}$ as $G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$.

With these defined, we present a fundamental result of General Relativity that will spur our discussion of gravitational waves.

Theorem 4 (Einstein's Equation)**.** *With a spacetime given by the metric $g_{\mu\nu}$, the following holds:*

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{5}$$

where G is Newton's Gravitational Constant and $T_{\mu\nu}$ is the Stress-Energy Tensor.

Remark 5. To interpret $T_{\mu\nu}$, we may view matter as a perfect fluid, i.e., one that is purely determined by its density ρ and pressure p. In such a case, the tensor becomes

$$T_{\mu\nu} = \begin{pmatrix} -\rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix}. \tag{6}$$

♣

In broad strokes, Einstein's equation states that the curvature of spacetime is related to the energy and momentum contained in the spacetime. This can be somewhat unintuitive but is a beautiful fact overall.

The proof for this comes purely from an experimental viewpoint. In particular, the detection of gravitational waves motivates this result because, as we shall soon see, they are a direct consequence of the Einstein Equation in the proper limit.

To derive the gravitational wave equation, we first choose the appropriate coordinates for the system. In this case, we choose the *harmonic coordinates,* and the corresponding gauge condition is $g^{\mu\nu}\Gamma^{\lambda}{}_{\mu\nu} = 0$. In the weak field limit, ($|h_{\mu\nu}| \ll 1$) this condition reduces to

$$\partial_\lambda h^{\lambda}{}_{\mu} = \frac{1}{2}\partial_\mu h. \tag{7}$$

Let us now define the *trace-reversed perturbation* $H_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$. This comes up multiple times in the calculation, so it is useful to give it a name. The name comes from:

$$\mathrm{Tr}\,(H_{\mu\nu}) = H^{\mu}{}_{\mu} = h^{\mu}{}_{\mu} - 2h = -h. \tag{8}$$

Using this definition, we find that the harmonic gauge condition simplifies further to $\partial_\mu H^{\mu}{}_{\nu} = 0$.

Plugging this into the Einstein Equations, we find that

$$\Box h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\Box h = -16\pi G T_{\mu\nu}$$

$$\implies \Box H_{\mu\nu} = -16\pi G T_{\mu\nu}. \tag{9}$$

In a vacuum, $T^{\mu\nu} = 0$ because there is clearly no density or pressure in a vacuum. Then, we obtain $\Box H_{\mu\nu} = 0$, which is a wave equation!

We then may solve this with a function of the form

$$H_{\mu\nu} = A_{\mu\nu} e^{ik_\alpha x^\alpha}, \tag{10}$$

where $A_{\mu\nu}$ is a constant rank 2 tensor and k_α is the *wave vector.* A good way to interpret $A_{\mu\nu}$ is as the initial conditions of the wave as $x_\alpha = 0$ implies that $H_{\mu\nu} = A_{\mu\nu}$.

Recall that in one dimension, wave equations can be written as $\psi \sim \exp\left(i(kx - \omega t)\right)$, which can be rewritten as $\psi \sim \exp\left(i(-\omega, k)\cdot(t, x)\right)$. Think of (ω, k) as the prototype wave vector and extend this to the number of dimensions necessary, replacing the dot product with the Minkowski version.

By plugging in the solution to the original wave equation, we find that $k_\alpha k^\alpha = 0$, meaning that the vector lies tangent to the worldline of a photon. This directly implies that gravitational waves move at the speed of light.

Our goal now is to reduce the number of parameters to 2, which is the usual amount for a plane wave. This section will be rather brief; for full detail, see sources such as [3] or others.

We first conclude from the harmonic gauge that $k_\mu A^{\mu\nu} = 0$, meaning that in a sense, $A^{\mu\nu}$ is orthogonal to k_μ. The formal term for this is that $A^{\mu\nu}$ is *transverse* to k_μ. Also, it is useful to make a coordinate change $x^\mu \mapsto x^\mu + \xi^\mu(x^\nu)$, and the original equation holds if we force $\Box\xi^\mu = 0$. Then, from solutions to this equation, we find a new matrix $A'_{\mu\nu}$ that is *traceless*, namely that $A'^{\mu}{}_{\mu} = 0$. As well, it is transverse to a constant four velocity u^ν. Under these two assumptions, we form the *transverse traceless (TT)* coordinates.

In the TT coordinates, the matrix A looks like

$$A'_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & A_{xx} & A_{xy} & 0 \\ 0 & A_{xy} & -A_{xx} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{11}$$

We are left with 2 degrees of freedom, as desired, so we are done. Note that this result also has further significance, as it implies that there are only two polarizations for gravitational waves.

While we have derived these only in the vacuum case, we will assume the full results in the coming sections.

IV. CRITICAL DIMENSION DERIVATION

We now take a turn into string theory to rigorously derive the dimension of spacetime for a quantum bosonic string in the light cone gauge. Before moving to the cosmic level, it is useful to consider quantum bosonic strings as a test case because we have much knowledge about these.

We first lock down some basic notation for the subject and give the intuition for the objects we are dealing with before we quantize everything.

Notation 6. For $X^\mu = X^\mu(\tau, \sigma)$ being the position function of a point on a string, we denote:

1. $\dot{X}^\mu = \partial_\tau X^\mu$,

2. $X^{\mu\prime} = \partial_\sigma X^\mu$.

With that settled, we recall the solutions to the wave equation $\ddot{X}^\mu - X^{\mu\prime\prime} = 0$. In terms of a Fourier Expansion, we may write

$$X^\mu(\sigma,\tau) = x_0^\mu + \sqrt{2\alpha'}\alpha_0^\mu \tau + i\sqrt{2\alpha'}\sum_{n\neq 0}\frac{1}{n}\alpha_n^\mu e^{-in\tau}\cos(n\sigma). \tag{12}$$

For a detailed derivation, see [8], Chapter 9. Unpacking this equation, we first let α' be the *slope parameter* with units $[\alpha'] = [E]^{-2}$. Furthermore, we define the coefficients $\{\alpha_n^\mu\}_{n\in\mathbb{N}}$ to be the *bosonic oscillator modes,* and more specifically, for a fixed $N \in \mathbb{Z}$, we say α_N^μ is the *Nth bosonic oscillator mode.*

It is of interest to simplify this sum, so for $n \geq 0$, we define the *Virasoro Modes*

$$L_n := \frac{1}{2}\sum_{p\in\mathbb{Z}} \alpha_{n-p}^I \alpha_p^I, \tag{13}$$

where α_z^I represents the transverse coordinates of α_z. In particular, for $n = 0$, we obtain

$$L_0 = 2\alpha' p^+ p^-, \tag{14}$$

where the latter 2 terms are the $+$ and $-$ coordinates for momentum in the light cone gauge. Note that for $A^\mu = (A^0, A^1, A^2, A^3)$, we define $A^\pm := \frac{A^0 \pm A^1}{\sqrt{2}}$.

With some work, one can show that in fact,

$$X^-(\sigma,\tau) = x_0^- + \frac{1}{p^+}L_0\tau - \frac{i}{p^+}\sum_{n\neq 0}\frac{1}{n}L_n e^{-in\tau}\cos(n\sigma), \tag{15}$$

thus the Virasoro modes are the Fourier modes of X^-.

We now quantize the theory presented, promoting classical variables to operator status. We thus send Virasoro modes to *Virasoro generators* and deal with them as an algebra with certain useful commutation properties. We formally present these now.

Definition 7. For $n \geq 0$, we define the *Virasoro generators* as

$$L_n := \frac{1}{2}\sum_{p\in\mathbb{Z}} \alpha_{n-p}^I \alpha_p^I;$$

$$\tilde{L}_n := \frac{1}{2}\sum_{p\in\mathbb{Z}} \tilde{\alpha}_{n-p}^I \tilde{\alpha}_p^I. \tag{16}$$

Here are some additional comments on the Virasoro generators:

- Splitting up X^μ into left and right-moving parts, we let $\{L_n\}$ be the right-moving Virasoro generators and $\{\tilde{L}_n\}$ be the left moving ones.
- To deal with the case of $n < 0$, we let $L_{-n} = L_n^\dagger$, where $\dagger$ represents the adjoint operator, and similar remarks hold for $\tilde{L}_{-n}$.

In general, we will only deal with the right-moving generators in this section. Here is a commutation identity and a couple corollaries that will prove useful.

Proposition 8. *Let D be the dimension of spacetime and $\{L_m\}$ be members of the quantized Virasoro algebra, defined previously. Then,*

$$[L_m, L_n] = i(m-n)L_{m+n} + \frac{D}{12}m(m^2-1)\delta_{m,-n}. \tag{17}$$

Corollary 9.

1. $L_1 L_{-1} = 2L_0 + L_{-1}L_1$.
2. $[L_1, L_2 + \gamma\,(L_{-1})^2] = 3L_1 + 2\gamma L_0 L_1 + 2\gamma L_1 L_0$, *for a fixed* $\gamma \in \mathbb{R}$.

We now give a couple of definitions that will be essential in our calculation.

Definition 10. In the context of the Virasoro algebra, we say that a state $|\psi\rangle$ is *physical* if $L_n|\psi\rangle = 0$ for all $n > 0$.

In practice, we need only check that $L_1|\psi\rangle = 0$ and $L_2|\psi\rangle = 0$ because given these, by the properties of the Virasoro algebra, we may readily verify that $L_n|\psi\rangle = 0$ also for $n \geq 3$.

Definition 11. A state $|\psi\rangle$ is called *spurious* if it satisfies the mass shell condition and is orthogonal to all other states. In symbols, this is: $(L_0 - a)|\psi\rangle = 0$ and for a physical state ϕ, $\langle\phi|\psi\rangle = 0$.

Note that the mass-shell condition is usually written as $p^2 + m^2 = 0$, but for our circumstances, the former equation is more desirable.

Here is an example of a spurious state to motivate the definition we will end up using. Let $\{|\chi_n\rangle\}$ be physical states with the property that $(L_0 - a + n)|\chi_n\rangle = 0$ and define

$$|\psi\rangle = \sum_{n\in\mathbb{N}} L_{-n}|\chi_n\rangle . \tag{18}$$

We may readily check that this has the desired properties. However, because of the commutation properties ($L_{-3} = [L_{-1}, L_{-2}]$ for example), we only need two terms, so we may write

$$|\psi\rangle = L_{-1}|\chi_{-1}\rangle + L_{-2}|\chi_{-2}\rangle . \tag{19}$$

We are also interested in our spurious states being physical, so we would like

$$\langle\psi|\psi\rangle = \sum_{n\in\mathbb{N}} \langle\chi_n| L_n |\chi_n\rangle = 0. \tag{20}$$

We will call such states *zero-norm states.* Without further ado, we now present the calculation for the spacetime dimension D in two parts.

A. Solving for a

Let $|\psi\rangle = L_{-1}|\chi\rangle$ be a zero-norm state such that

$$(L_0 - a + 1)|\chi\rangle = 0\,;\; L_m|\chi\rangle = 0 \tag{21}$$

for $\mathbb{Z} \ni m > 0$. Since $|\psi\rangle$ is a zero-norm state, it is also physical, so $L_m|\psi\rangle = 0$ for all $m > 0$. We choose $m = 1$ to exploit Identity 1 in Corollary 9, implying

$$L_1|\psi\rangle = L_1 L_{-1}|\chi\rangle = (2L_0 + L_{-1}L_1)|\chi\rangle$$

$$= 2(a-1)|\chi\rangle = 0, \text{ so } a = 1. \tag{22}$$

Note that some authors ([8]) will write $a = -1$, but they set up the problem along the lines of $(L_0 + a + 1)|\chi\rangle = 0$, so everyone is right.

B. Solving for D

Let $|\psi\rangle = \left(L_{-2} + \gamma L_{-1}^2\right)|\chi\rangle$ be a zero-norm state for a particular γ such that

$$(L_0 + 1)|\psi\rangle = 0 = L_m|\psi\rangle , \tag{23}$$

for $\mathbb{Z} \ni m > 0$.

Since $|\psi\rangle$ is physical, we deduce that $L_1|\psi\rangle = L_2|\psi\rangle = 0$. Thus, we may obtain

$$L_1|\psi\rangle = L_1\left(L_{-2} + \gamma L_{-1}^2\right)|\chi\rangle \tag{24}$$

and use Identity 2 in Corollary 9 to write

$$((3 - 2\gamma)L_{-1} + 4\gamma L_0 L_{-1}) = 0. \tag{25}$$

The second term vanishes with some algebra that relies on (23), and the first term vanishes by setting $\gamma = 3/2$. We therefore fix $\gamma = 3/2$ for the remainder of the calculation.

We now force $L_2|\psi\rangle = 0$ and use the identity

$$\left[L_2, L_2 + \frac{3}{2}L_{-1}^2\right] = 13L_0 + 9L_{-1}L_1 + \frac{D}{2}, \tag{26}$$

which can be derived from Proposition 8. We then plug in

$$L_2|\psi\rangle = L_2\left(\frac{3}{2}L_{-1}^2\right)|\chi\rangle = \left(-13 + \frac{D}{2}\right)|\chi\rangle = 0,$$

forcing $D = 26$ as desired.

We will now move onto research circa 2008 that deals with gravitational waves, strings, and extra dimensions in order to further understand the current nature of spacetime.

V. MODERN RESEARCH

In this section, we will explore the formation and decay of cosmic superstrings. As an application, we give intuition for the gravitational wave background from cosmic strings.

A. Cosmic Superstrings and String Networks

So far, the scientific community has found little evidence for cosmic strings to exist, if not contradictory evidence proving they do not exist. More precisely, they are

"inconsistent with the emerging measurements of fluctuations in the microwave background" [6]. To move closer to reality, we voyage into *superstring theory*, which conjectures that superstrings exist in an $N-$dimensional spacetime, but we do not know N.

We will now consider cosmic superstring networks, their behavior, and their gravitational decay patterns. Note that all strings referred to here on out will be superstrings.

Classically, networks of cosmic superstrings form via $D3/\overline{D3}$ processes when there is a broken $U(1)$ symmetry. Note that $U(n)$ denotes the unitary $n\times n$ matrices, and it is useful to think of $U(1)$ as the "circle group" of complex numbers with absolute value 1.

In the case of a single string of scale greater than the horizon length, it forms as a *(unweighted) random walk*, i.e. that at any point in time, the string is equally likely to grow in every direction. This may cause the string to form kinks, cusps, or loops, defined later. Shorter strings tend to straighten out over time.

Modeling string networks is still an evolving field as it is highly nonlinear and highly model-dependent [7]. On the micro scale, when two strings collide, there are two cases:

1. The two strings detach and reattach in a new way. This happens with probability P.

2. The two strings pass through each other, remaining otherwise unchanged. This happens with probability $1-P$.

P is usually nearly 1 for gauge theory solitons, and P is exactly 1 for adiabatic collisions between gauge theory solitons as it is "energetically favorable" [7]. For graphical intuition, see Figure 1. As well, Figure 2 shows a simulated network of cosmic strings.

We now discuss various types of phenomena in the formation of strings.

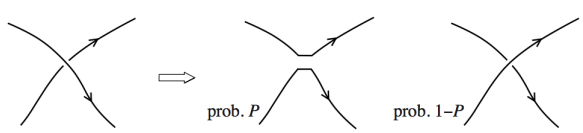

FIG. 1: A visual evolution of colliding strings, adapted from [7], Figure 1.

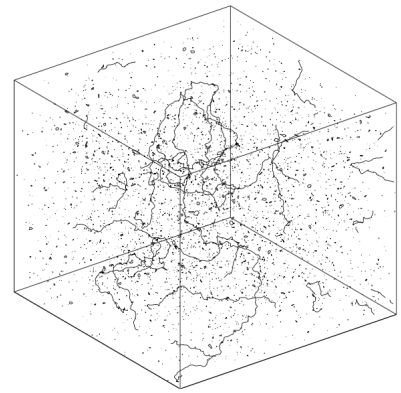

FIG. 2: A computer simulation of a string network, adapted from [7], Figure 2.

Definition 12. Let $X^\mu(\tau,\sigma)$ be a parameterization of the string position.

1. A *kink* occurs at $(\tau,\sigma)=(\tau_0,\sigma_0)$ when there is a discontinuity in $X^{\mu\prime}$ at (τ_0,σ_0). This appears as a sharp turn on the string.

2. A *cusp* occurs at (τ,σ) when $\dot{X}^\mu_+(\tau,\sigma)=\dot{X}^\mu_-(\tau,\sigma)$. Visually, this is similar to what happens when you pull sewing thread together to pass it through a needle.

3. A *loop* occurs if the string passes over itself, more precisely: for a fixed τ_0 and two distinct σ_1,σ_2, $X^\mu(\tau_0,\sigma_1)=X^\mu(\tau_0,\sigma_2)$ where

$$X^\mu\left(\tau_0,[\sigma_1,\Sigma]\right)\neq X^\mu\left(\tau_0,[\Sigma,\sigma_2]\right) \tag{27}$$

for all $\Sigma\in(\sigma_1,\sigma_2)$. A loop in the string looks like what you think it looks like.

Remark 13. Some notes on Item 3: We first define the interval notation given in 27 as the image of X^μ under an interval in σ. More precisely, if we fix τ and give an interval A,

$$X^\mu\left(\tau,A\right)=\{X^\mu(\tau,\sigma)\,|\,\sigma\in A\} \tag{28}$$

The motivation for including the condition (27) in the first place is to force our "loops" to not be a large cusp with sections of the string glued together. ♣

Typically, when cusps and loops form, there is a large release of energy. In the case of a cusp, a good metaphor to think of is the crack of a whip. When loops form,

they usually break off from the original string and decay, emitting gravitational radiation. These events cause the gravitational wave background we will investigate soon.

B. Motion and Extra Dimensions

As one may expect, in higher dimensions, there is a higher conceptual framework necessary as there is so much more room to work. Thus, to aid in grappling with higher-dimensional thinking, one may wish to consult [1] for a pedagogical mathematical treatment. In string theory, we postulate the existence of large extra dimensions (LED's), causing a reduced probability for cusps to form and an overall round-off of cusps [6]. Therefore, the rate of gravitational wave production and the power of each wave decreases, making detection harder. This is also heavily dependent on the number of extra dimensions because as shown in [1], adding even 1 or 2 dimensions leads to a large increase in the amount of "room" there is to work with.

Since there are fewer cusp events (CE's) in higher dimensions, we are interested in what happens when a string almost forms a cusp. We call these *Near-Cusp Events,* or NCE's. To formalize this, if

$$\left|\dot{X}^{\mu}_{+} - \dot{X}^{\mu}_{-}\right| < 2\Delta \ll 1 \tag{29}$$

at some (τ, σ), we say that there is a NCE with parameter Δ. The purpose of Δ is to tell us how close to a CE the event is. In particular, if $\Delta = 0$, the event was indeed a CE.

In test cases, we find that $\mathcal{N}(\Delta)$, the number of NCE's in a given loop with parameter Δ, is $\mathcal{N}(\Delta) = O(\Delta^n)$ for some n. In another article, the authors of [6] find n to be the number of extra dimensions in spacetime, which should be intuitive. Letting $C(\Delta)$ be the local probability density of the network, we find that $C(\Delta) = n\Delta^{n-1}$ and is proportional to the derivative of $\mathcal{N}$. We use this to find the rate of Gravitational Wave events and integrate to find the net effect of NCE's.

This helps us understand the nature of the gravitational waves we hope to detect, and (I think) helped us in finding appropriate ranges for LIGO and LISA. For diagrams comparing these hypothesized results with LIGO and LISA curves, see page 3 of [6].

The most interesting part of this research is that the dependence on the dimension n. Namely, we might be able to use these signals to explicitly determine n and confirm the dimensionality of spacetime.

C. Gravitational Wave Spectrum

We now turn to the stochastic gravitational wave background and give concluding remarks. Instead of viewing the decay of strings as a micro-level process, we view the gravitational wave background generated by the decay of all cosmic strings in the universe.

First, we define the dimensionless quantity Ω as

$$\Omega := \frac{1}{\rho_c}\frac{d\rho_g}{d\ln f}. \tag{30}$$

Here,

- ρ_g is the energy density of the gravitational waves.
- ρ_c is the critical density.
- f is the frequency in units of the critical density.

We use Ω to measure the energy density of the gravitational waves as a function of $\ln f$. An interesting fact is that Ω is proportional to the root mean square strain on spacetime caused by the gravitational wave background. Some additional useful quantities are the string tension in Planck units $G\mu$, the loop size over the Hubble radius α, and the number of strings created at a certain time in a Hubble volume $\frac{N_t}{\alpha}$. Figure 3 shows an energy density per frequency plot with fixed α, γ, N_t. for more plots that vary other parameters, see [4], pages 7-12.

To conclude, it is continually baffling how important LIGO's discovery of gravitational waves is. While verifying Einstein is a feat in itself, the real importance is where we can go with the knowledge that gravitational waves were the correct model. In general, the discovery allows us to put theory into practice and go to work on discovering the true nature and dimensionality of the universe.

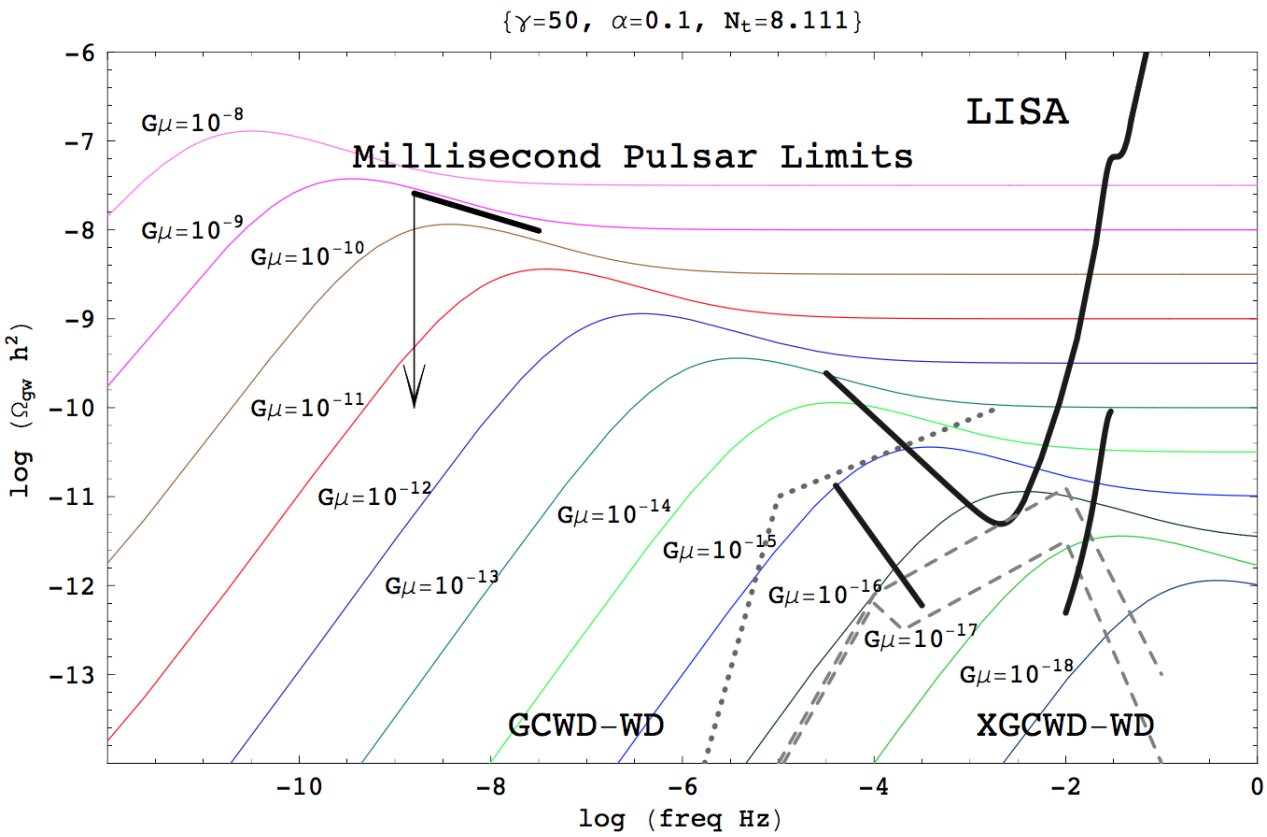

FIG. 3: A log-log plot of the gravitational wave energy density versus the frequency of the wave.

[1] 3Blue1Brown. *A Trick to Visualizing Higher Dimensions.* `https://www.youtube.com/watch?v=zwAD6dRSVyI` (9:10)

[2] Sean Carroll. *A No-Nonsense Introduction to General Relativity.* `https://preposterousuniverse.com/wp-content/uploads/2015/08/grtinypdf.pdf`

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